08 Hardware Metering

Engr 399/599: Hardware Security

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Adapted from: Mark Tehranipoor of University of Florida

Course Website

engr599.github.io

Write that down!

Agenda

Last time:
- PUFs!

This time:
- Project 2 Assigned Next Week!
- Review PUFs
- TRNG
- Metering
- Watermarking?

Next Time:
- Finish Watermarking
- Project 2 in class work.
- Practice Exam Released
    - Completion earns points for exam!

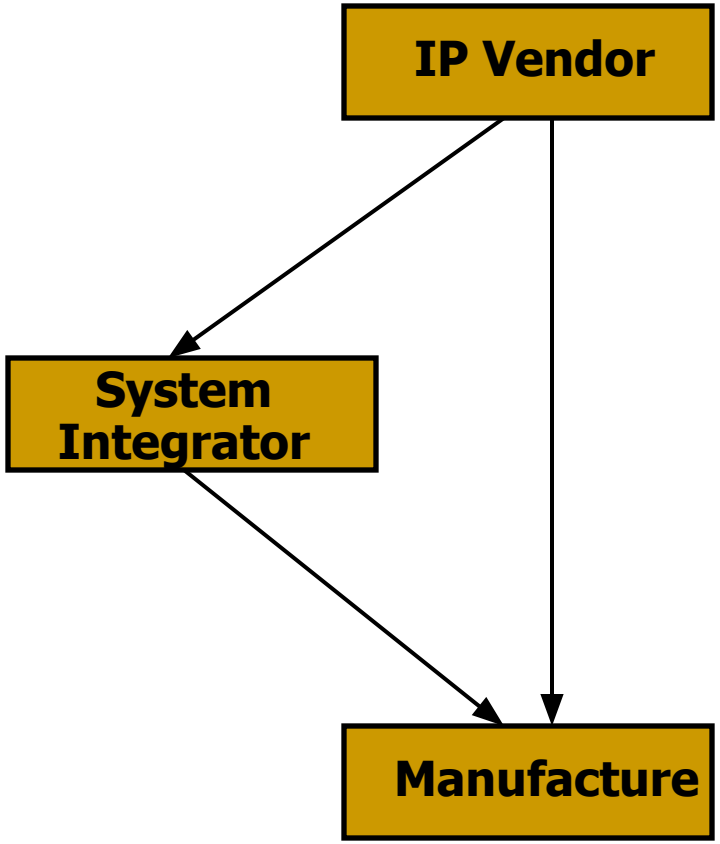
Project 2: Obfuscation & Watermarking

https://github.com/ENGR599/P2_Obfuscation/tree/2024_update

Review in class today.

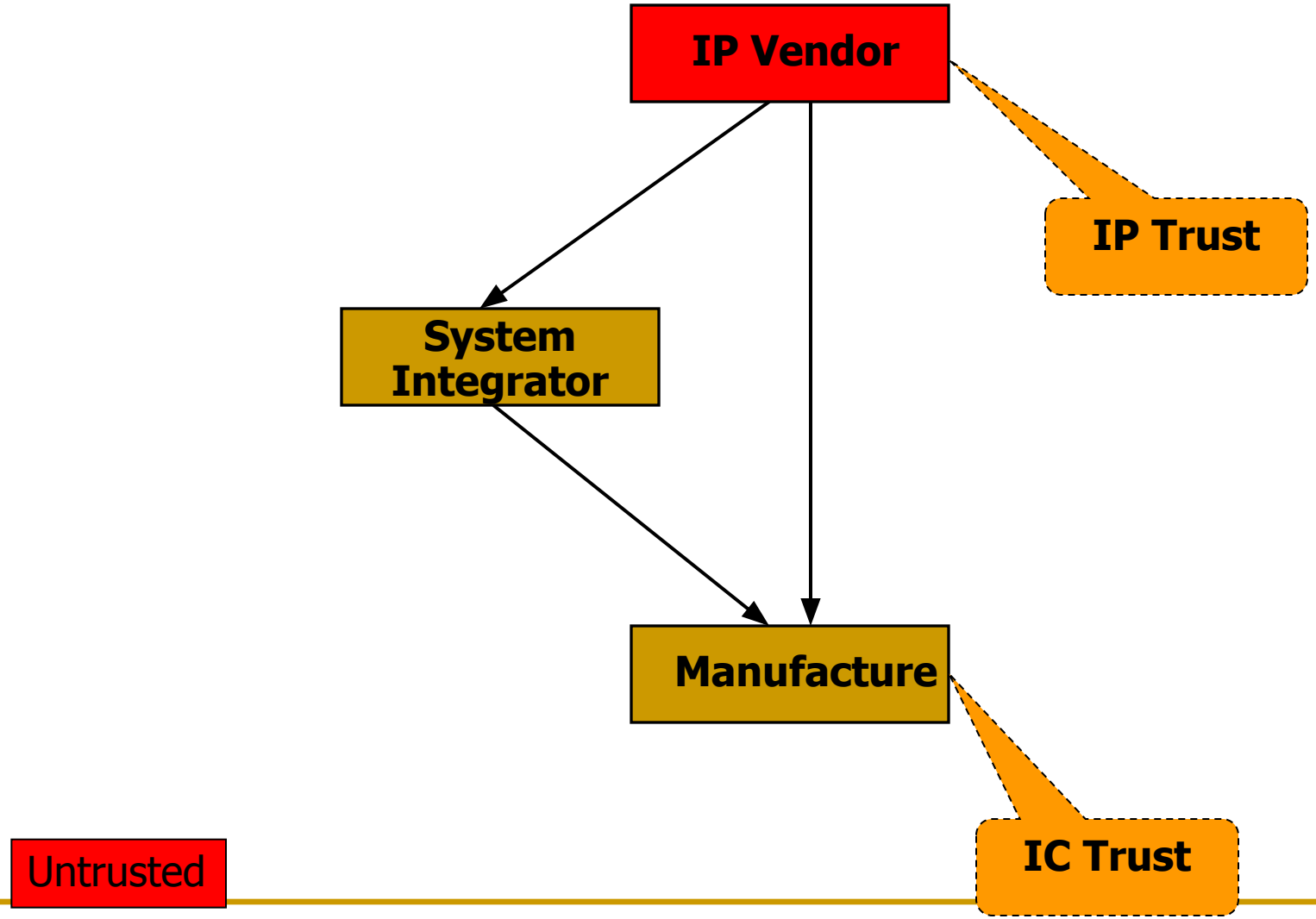
Needs a small update - ‘officially’ assigned Wednesday.

HW Threats

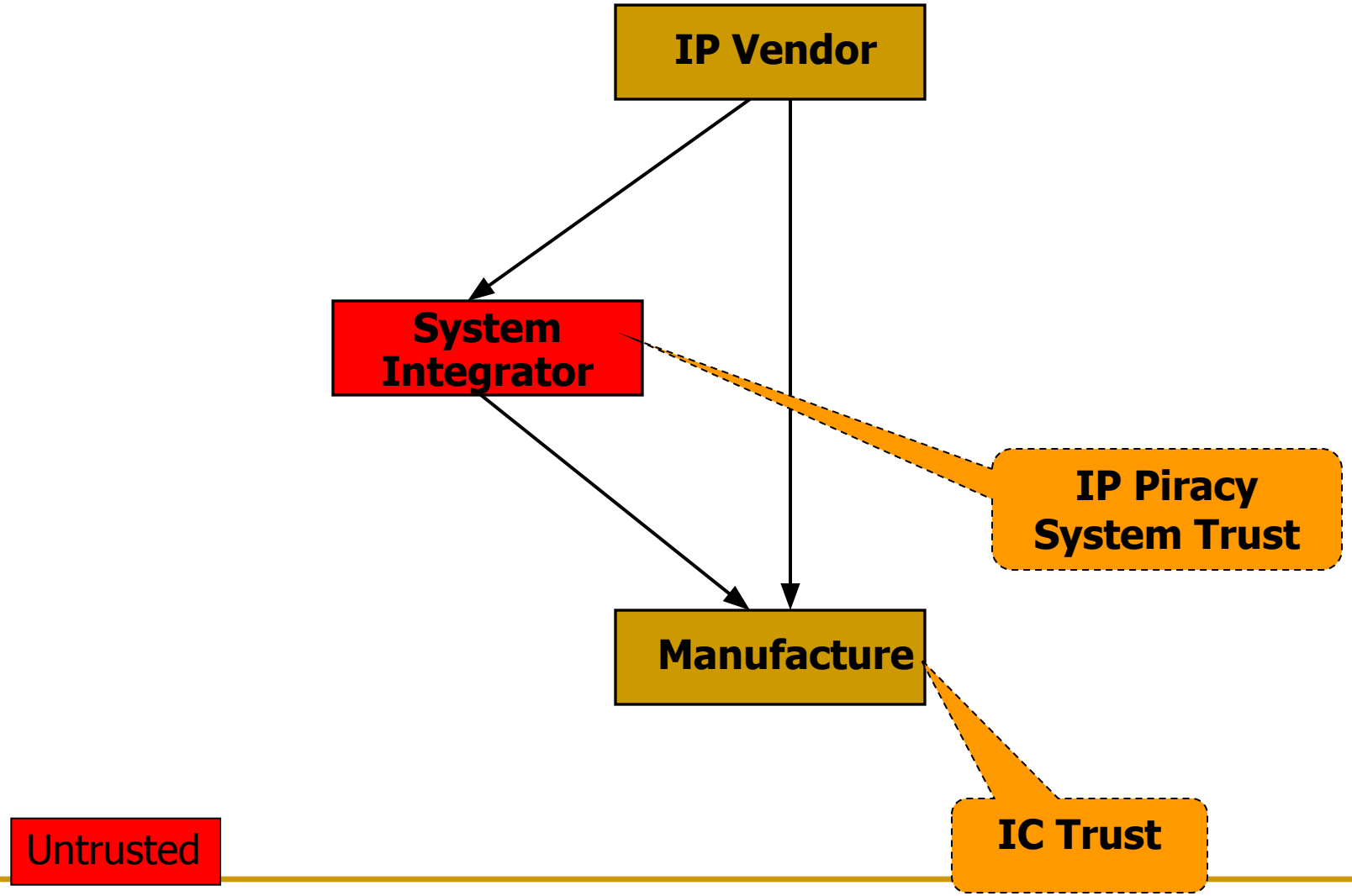


Any of these steps can be untrusted

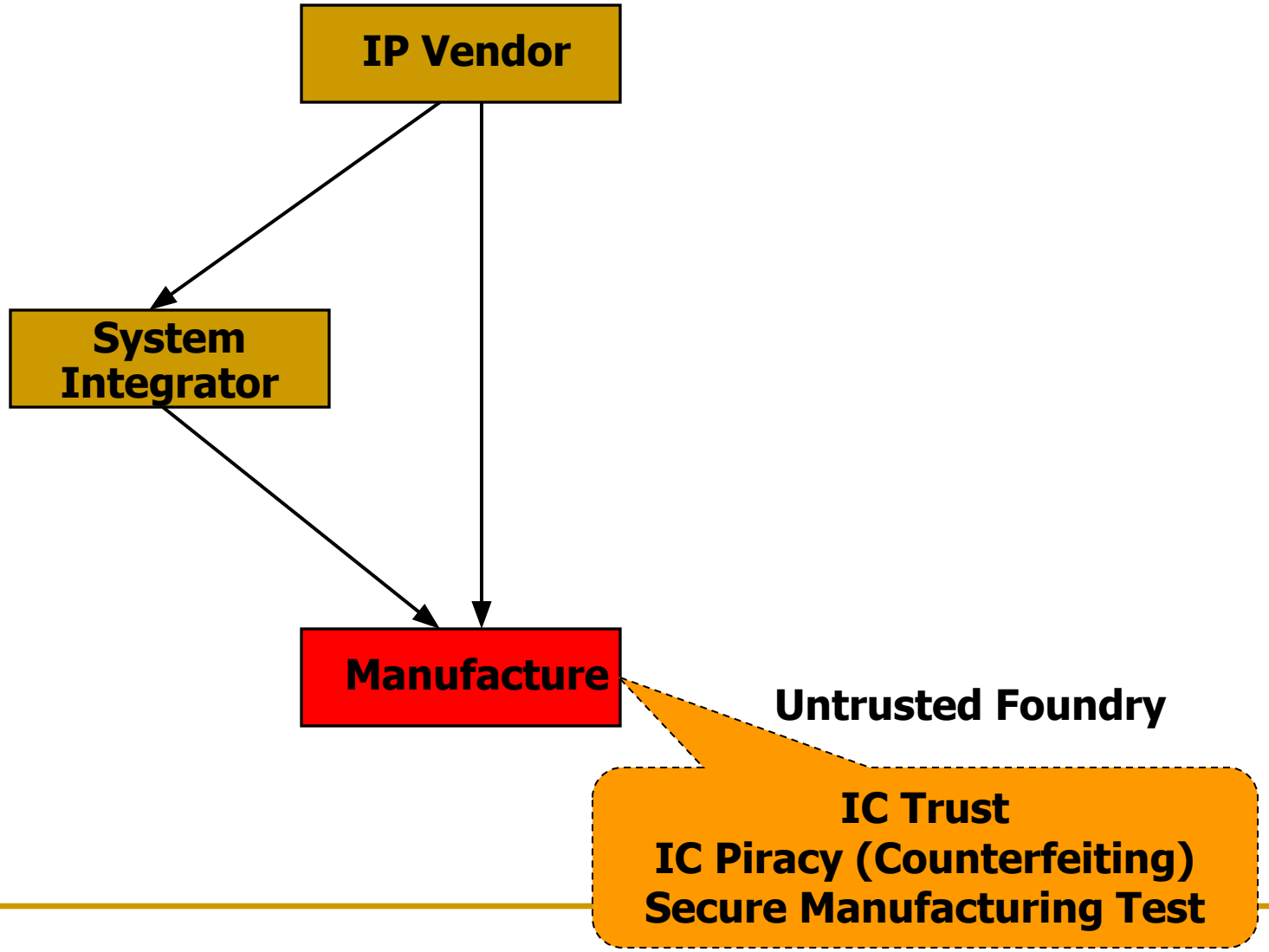
HW Threats


IP Vendor
IP Trust
System Integrator
Manufacture
IC Trust
Untrusted

HW Threats

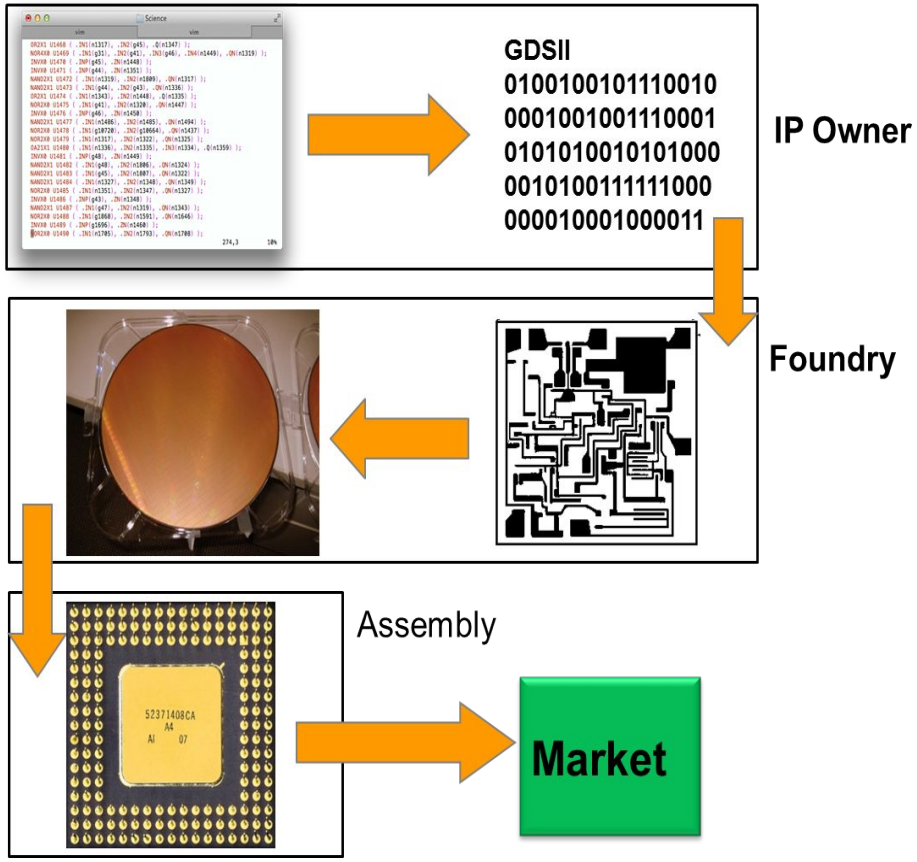

IP Vendor
System Integrator
IP Piracy
System Trust
Manufacture
IC Trust
Untrusted

HW Threats



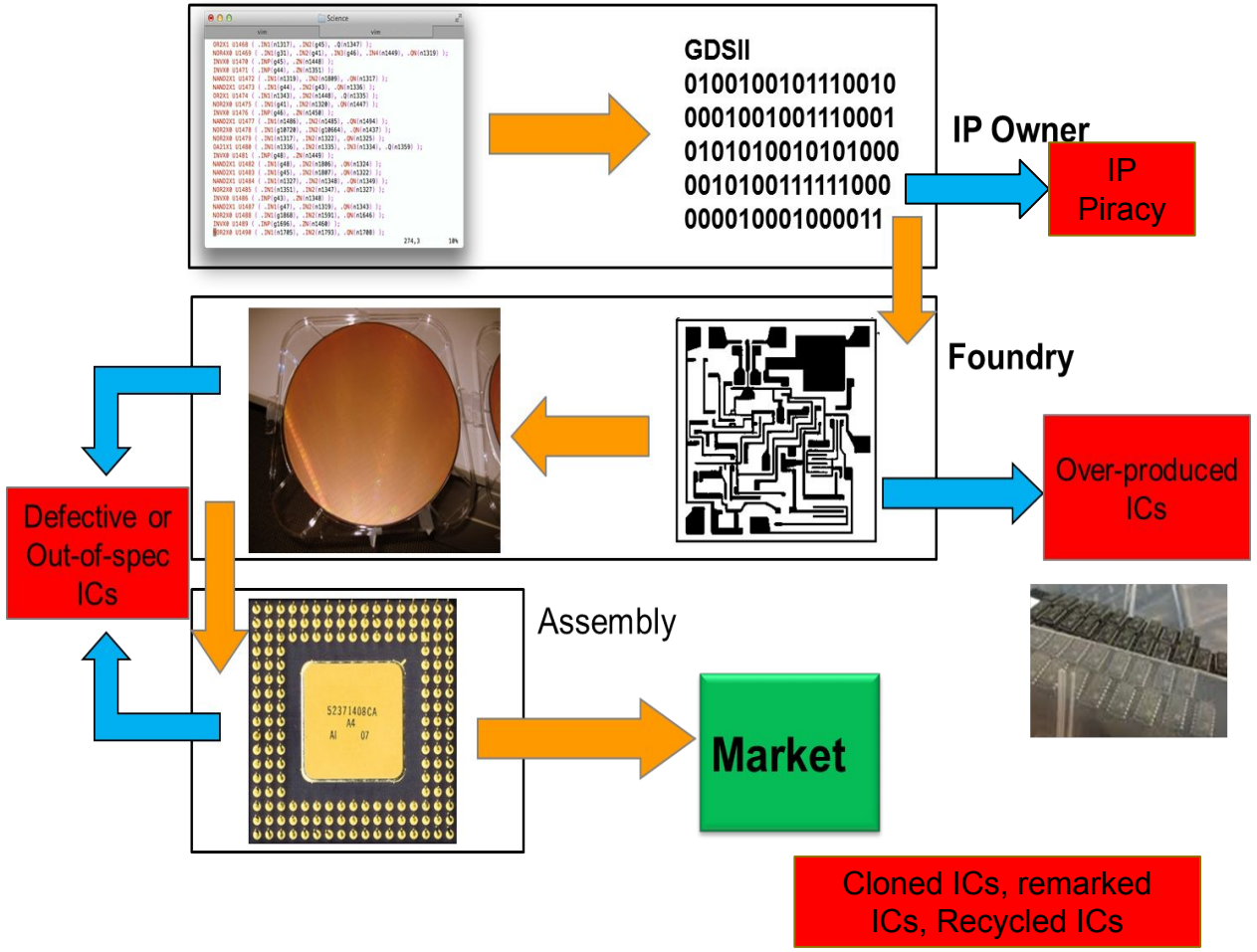
Untrusted

Chip Production Flow



- Little communication between IP Owner and Foundry.
- Foundry is trusted with full design.
- Responsible for production of requested amount of chips.
- IP holder provides foundry/assembly with all **test patterns** and **responses**.

Chip Production Flow

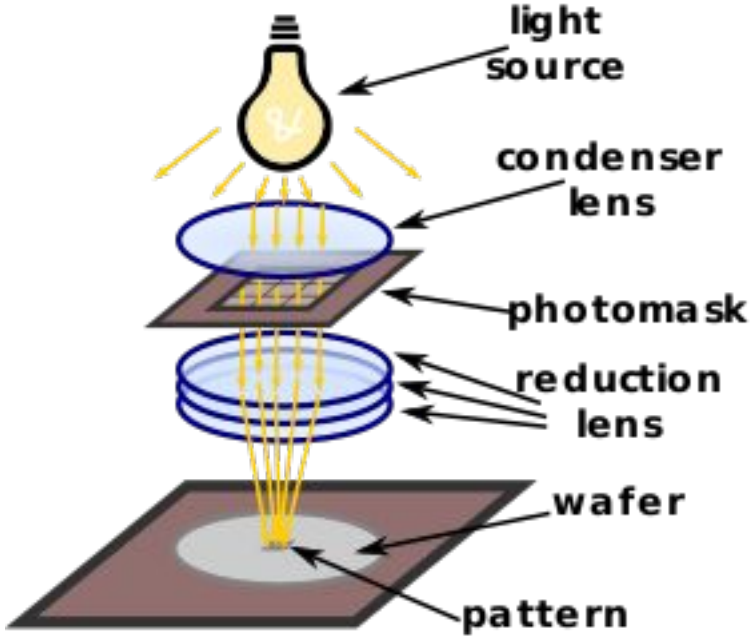


- Foundry looks for its own profit.
- Once mask is produced, producing IC's is simple and cheap.
- Lack of communication makes it **difficult for owner to track produced chips.**

Photomask ("Mask") is the expensive part

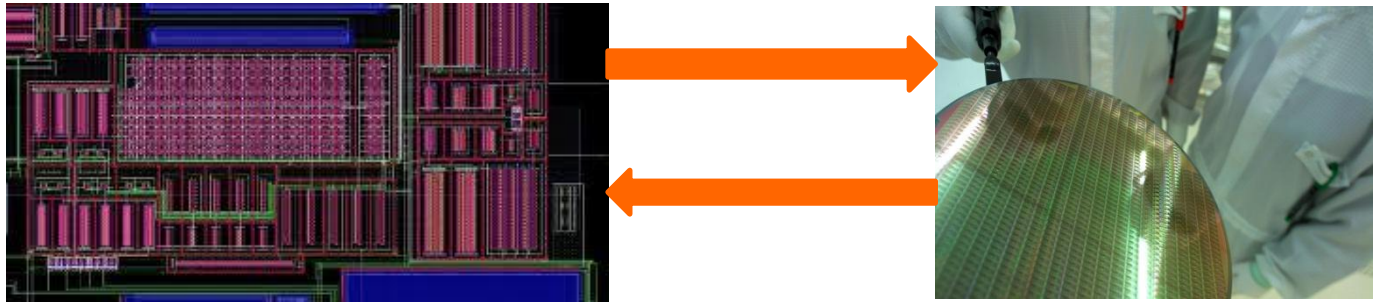
How much is a set of masks in today's world?

What are the costs that go into mask development?

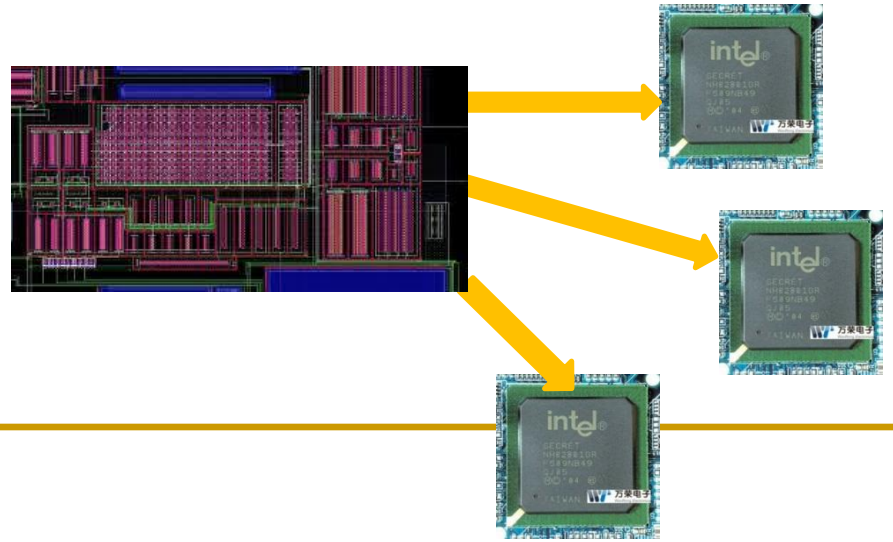


Need for Hardware Metering

- Need for better communication between IP Owner and foundry/assembly.



- Need for IP Owner to be able to track produced chips.

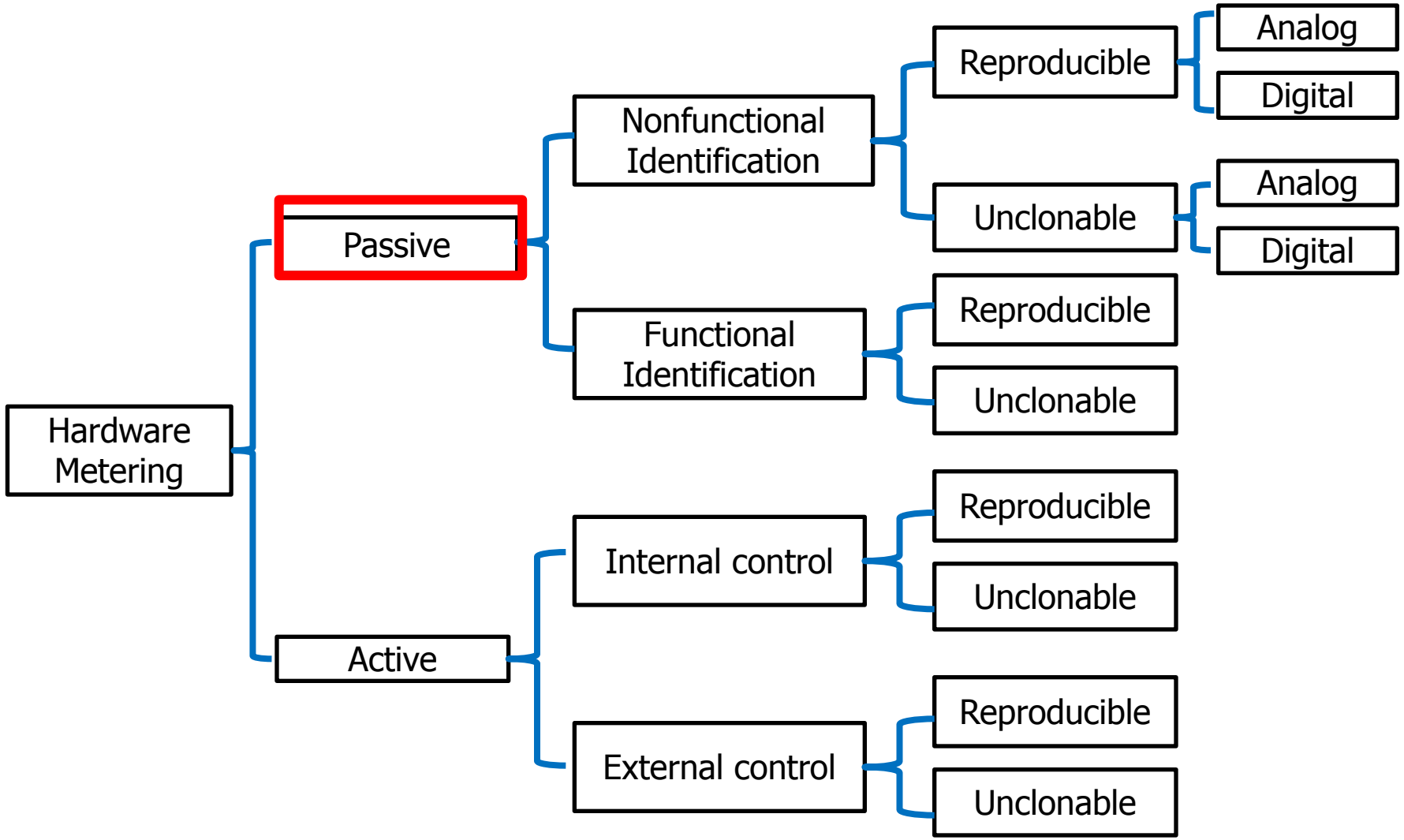


Electronic Chip ID (ECID)

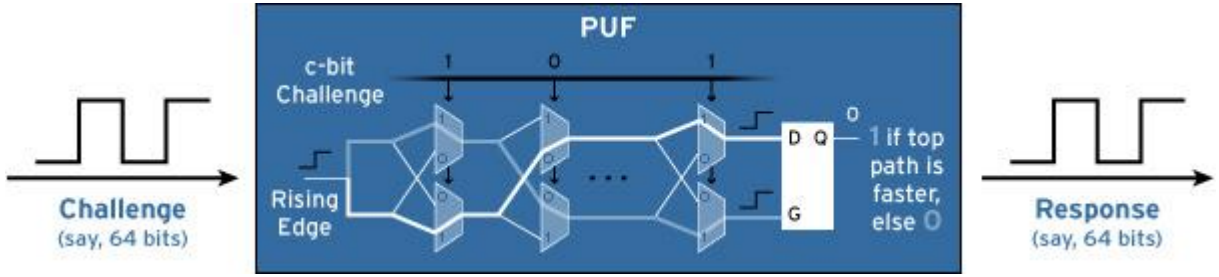
Hardware Metering

- **Hardware metering (IC metering):**
  - Set of security protocols that enable IP owners to achieve post-fabrication control over their ICs
  - Methods attempt to **uniquely tag each chip** to facilitate tracing them
  - Two main methods:
    - **Active metering**
    - **Passive metering**
- **Could be applicable to PCBs, e.g., IoTs**

Taxonomy of Metering Methods

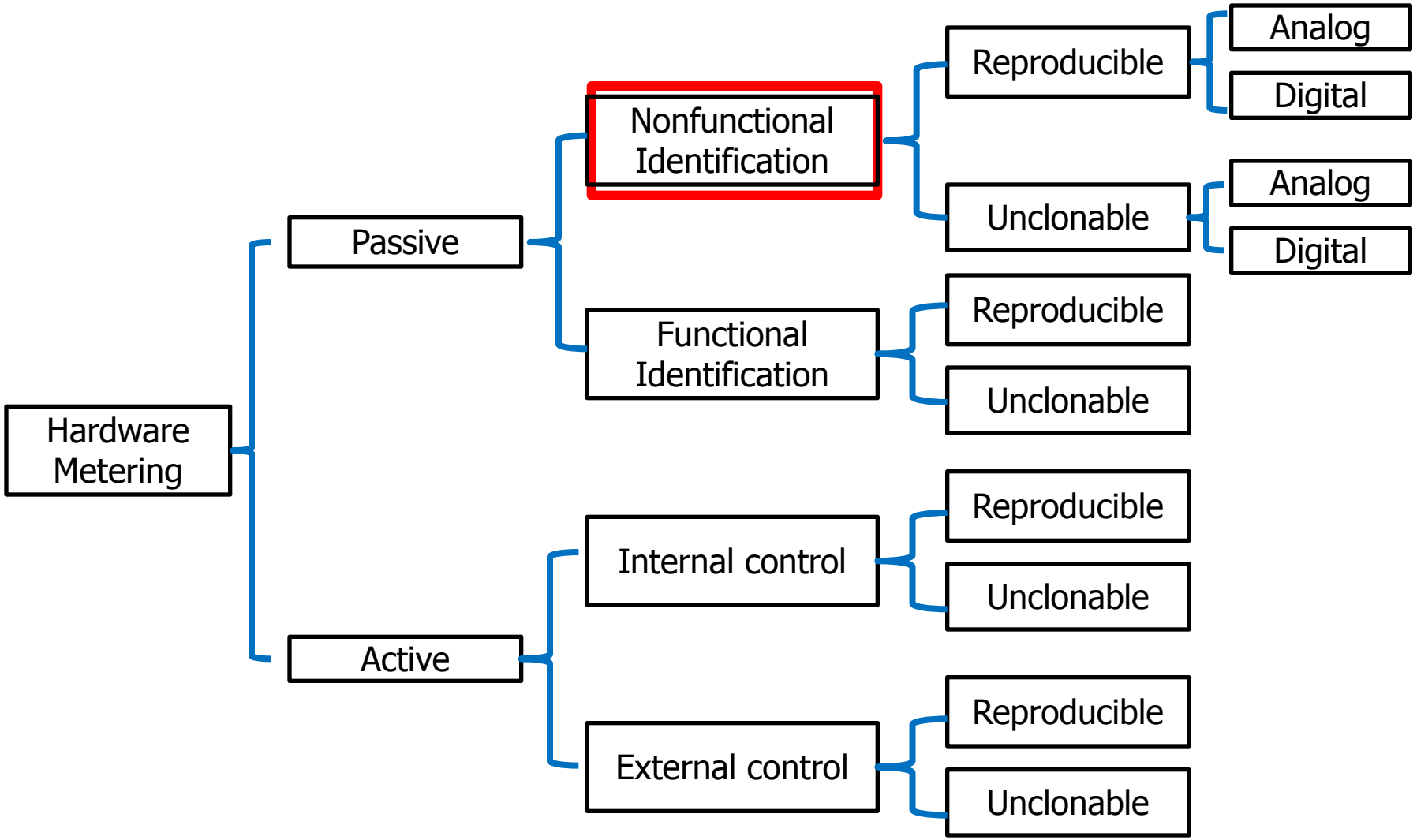


Passive Metering



- ICs can be **passively monitored**.
- Can be achieved by physically identifying:
  - Serial numbers on chips
  - Storing unique identifiers in memory. These are called Nonfunctional Identification
    - E.g., Electronic Chip ID (ECID)
- Tagging an IC's functionality: Functional Identification

Taxonomy of Metering Methods



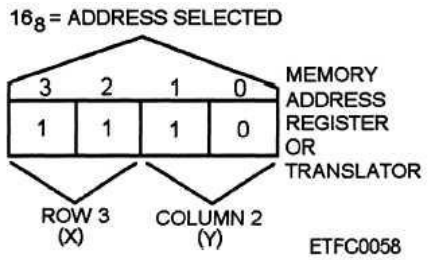
Nonfunctional Identification

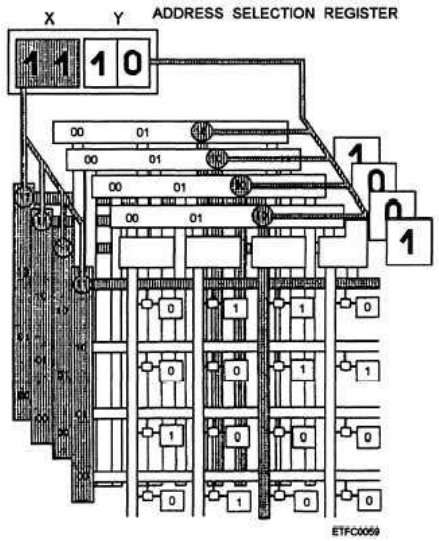
- Unique ID is separate from the chip's functionality.

- Vulnerable to cloning and/or removal.
  - Once chip is tagged, foundry can copy same tag on other chips or simply remove tag so chip cannot be traced.
- Possible to overproduce.
  - Foundry can produce multiple chips with same tag.
  - Out of millions of chips, probability of finding two matching tags is small.
- Two main types:
  - Reproducible
  - Unclonable

Nonfunctional Identification: Reproducible Identifiers

- Unique ID's are stored on the chip package, on die, or in a memory on-chip.
- Examples:
  - Indented serial numbers
  - Digitally stored serial numbers
- Advantages:
  - Do not depend on randomness
  - Easy to track / identify.
- Disadvantages:
  - Easy to clone/modify
  - Easy to overproduce





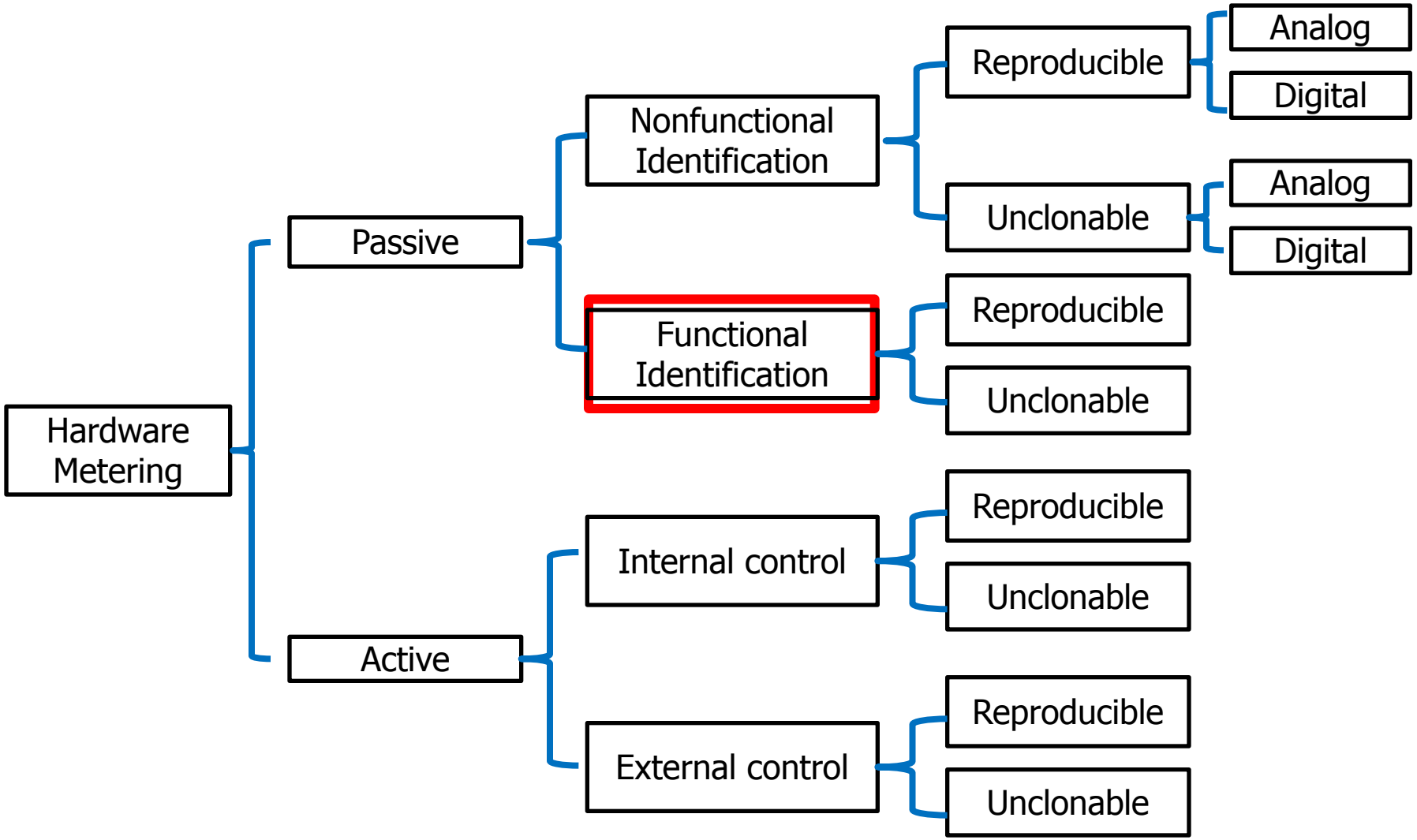
Nonfunctional Identification: Unclonable Identifiers

- Uses random process variations in silicon to generate random unique numbers called fingerprints.
- If additional logic is needed to generate these value, the method is said to be extrinsic.
- If no additional logic is needed, the method is called intrinsic.
- Advantages:
  - Values cannot be reproduced due to randomness in process variations
- Disadvantages:
  - Foundry could overproduce ICs without knowledge of IP owner
    - i.e., these methods do not prevent counterfeiting. The over-produced chip can be detected if IP owner gets his/her hands on those chips by comparing the identifier on the chip with his/her database

Unclonable Identifiers

- **Extrinsic methods:**
  - Require additional logic such as PUF (Physical Unclonable Function) or ICID
  - ICID
    - Threshold mismatches in array of transistors incurred different currents and therefore unique random numbers.
  - PUFs
    - Series of ring oscillators (ROs) generate random value due to process variations.
- **Intrinsic methods:**
  - Unique identification if external test vectors can be applied.
  - Uses IC **leakage**, **power**, **timing**, and **path signatures** (unique due to process variations).
  - Does not need additional logic and can be readily used on existing designs

Taxonomy of Metering Methods



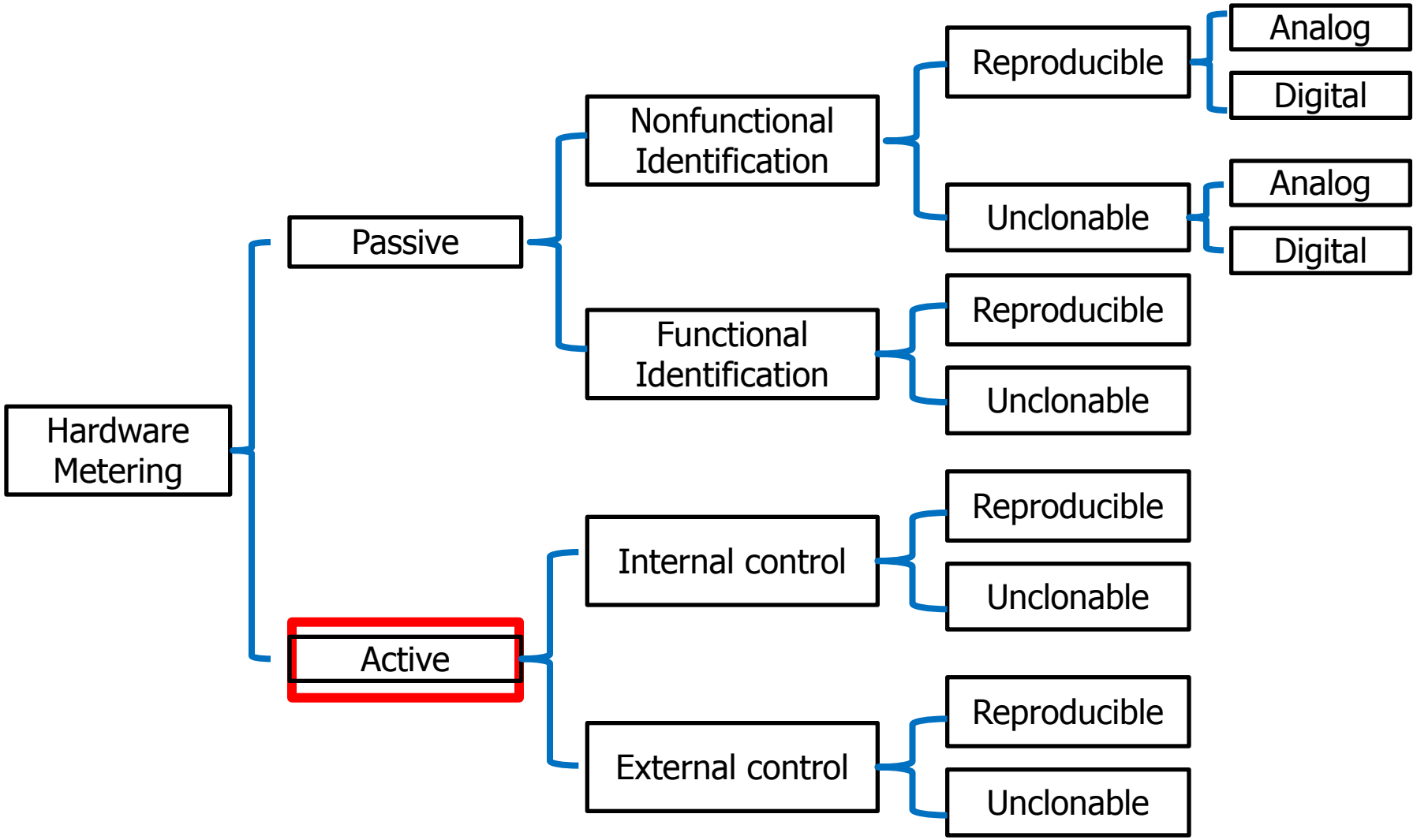
Functional Metering

- Identifiers linked to chip's internal functional details during synthesis.
- Each chip's function gets a unique signature.
  - E.g., additional states added to generate same output
- Function unchanged from input to output
- Internal transactions unique to each chip
- Challenge in fabricating ICs with different paths from same mask.

Functional Metering

- One method is fabricating chips from same mask and maintaining one programmable path.
  - E.g., Datapath could be programmed post-silicon.
  - IP Owner provides correct input/key combination to foundry to program chip post-silicon.
- Additional work proposes adding redundant states.
  - Programmable read logic enables selecting correct permutation for a control sequence.
- Drawbacks:
  - Testing such circuitry provides low coverage because the actual functionality of the chip is hidden during the test process by foundry and assembly
  - It requires the chip to go back to a trusted facility to be activated.

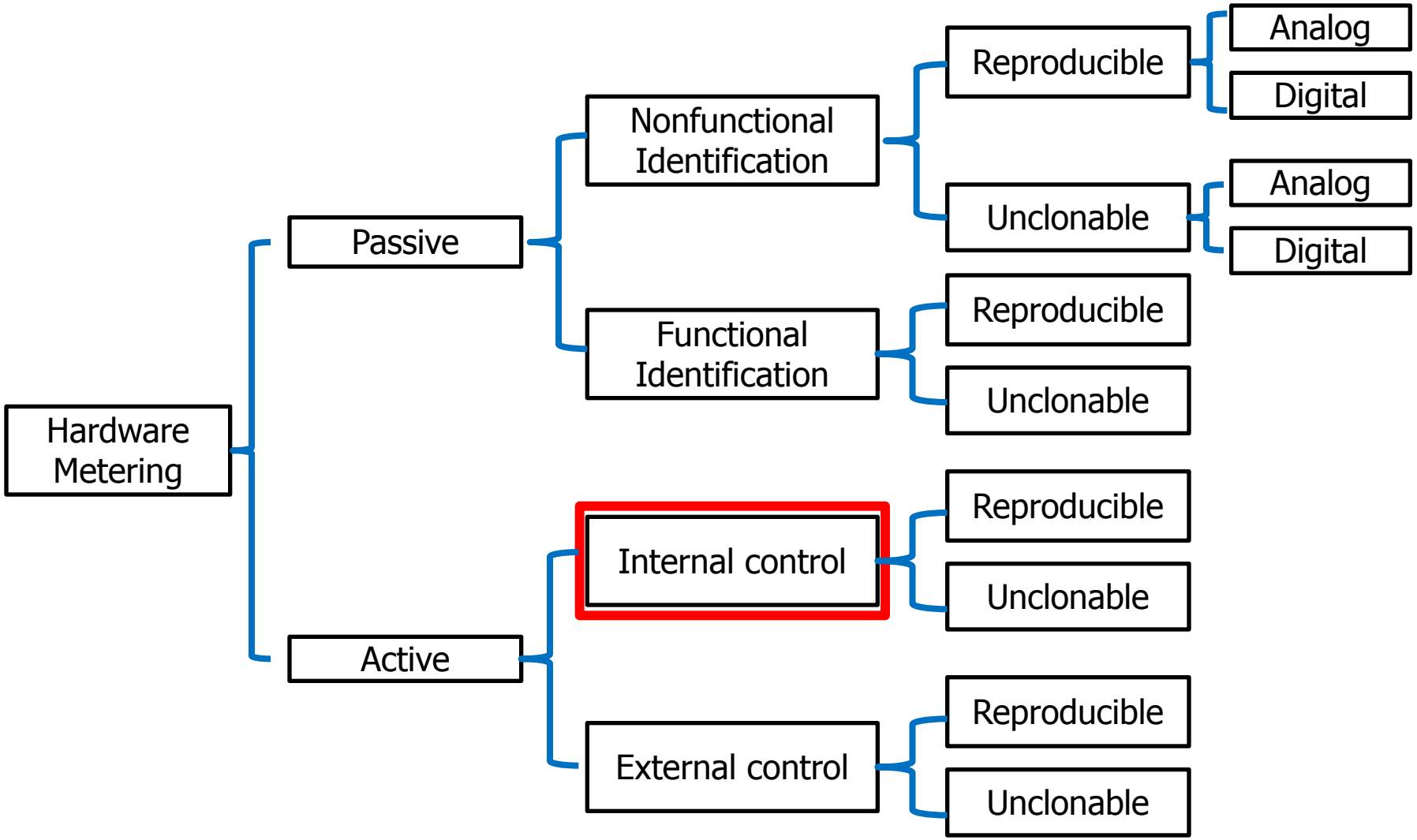
Taxonomy of Metering Methods



Active Metering

- Provides active way for designer to enable, control, or disable IC.
- Unlike passive metering, active metering requires ***communication between design house (IP owner) and foundry***.
- Two types:
  - Internal
  - External

Taxonomy of Metering Methods



Internal (Integrated) Active Metering

- Hides states and transition in the design that can only be accessed by designer.
- Locks are embedded within structure of computation model in hardware design in form of FSM.
- Adding additional states or duplicating certain states in FSM adds ability for designer to decide which datapath (sequence of states) to use post-silicon.
  - Since states are added, specific combinations are needed to bring FSM to correct output. Only IP owner knows such combination.

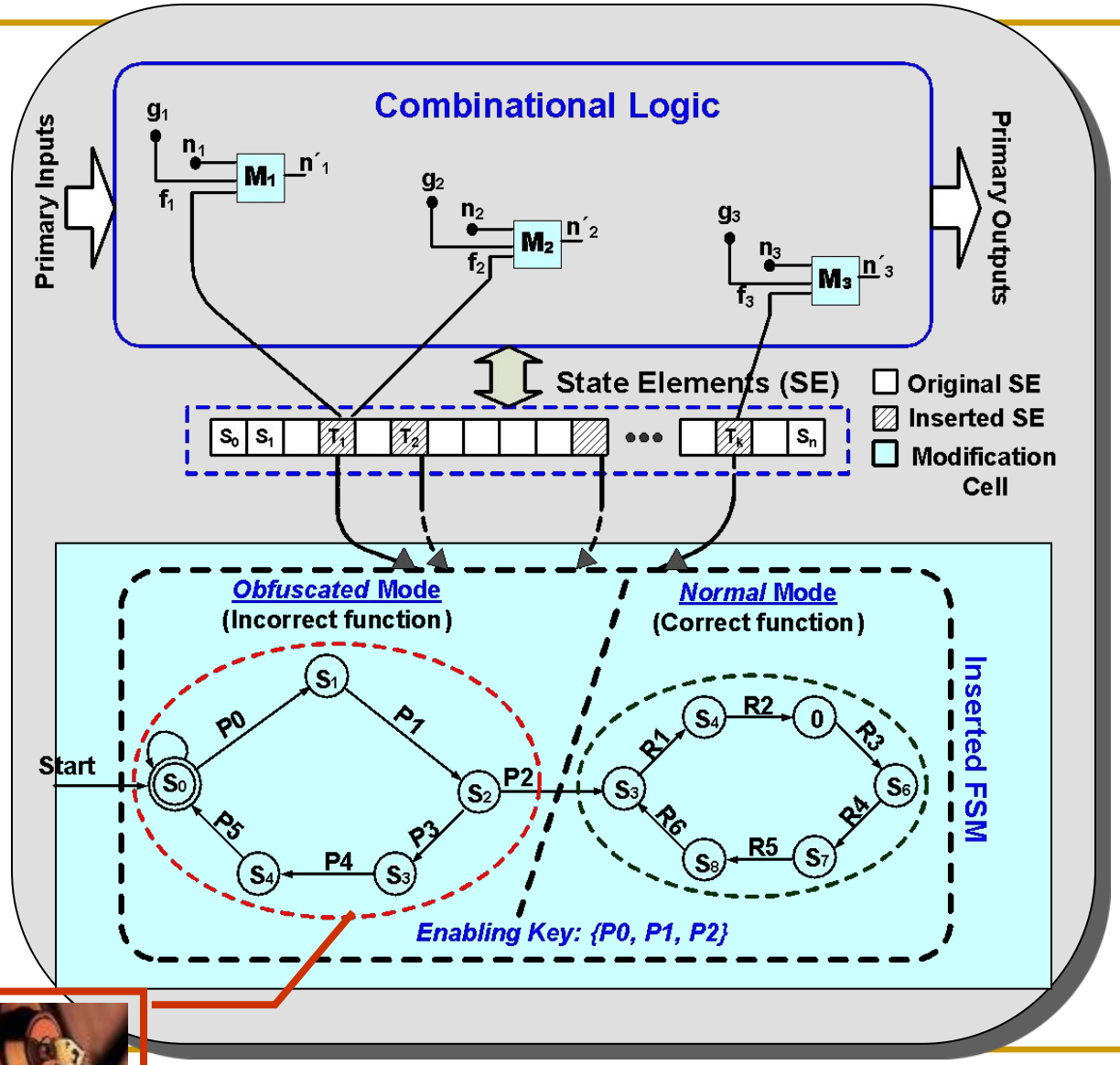
State Space Obfuscation

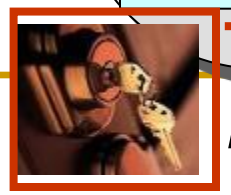
Basic Idea:

- A locking approach where normal behavior is *enabled* only upon appn. of a key
- Provable robustness

Key Innovations:

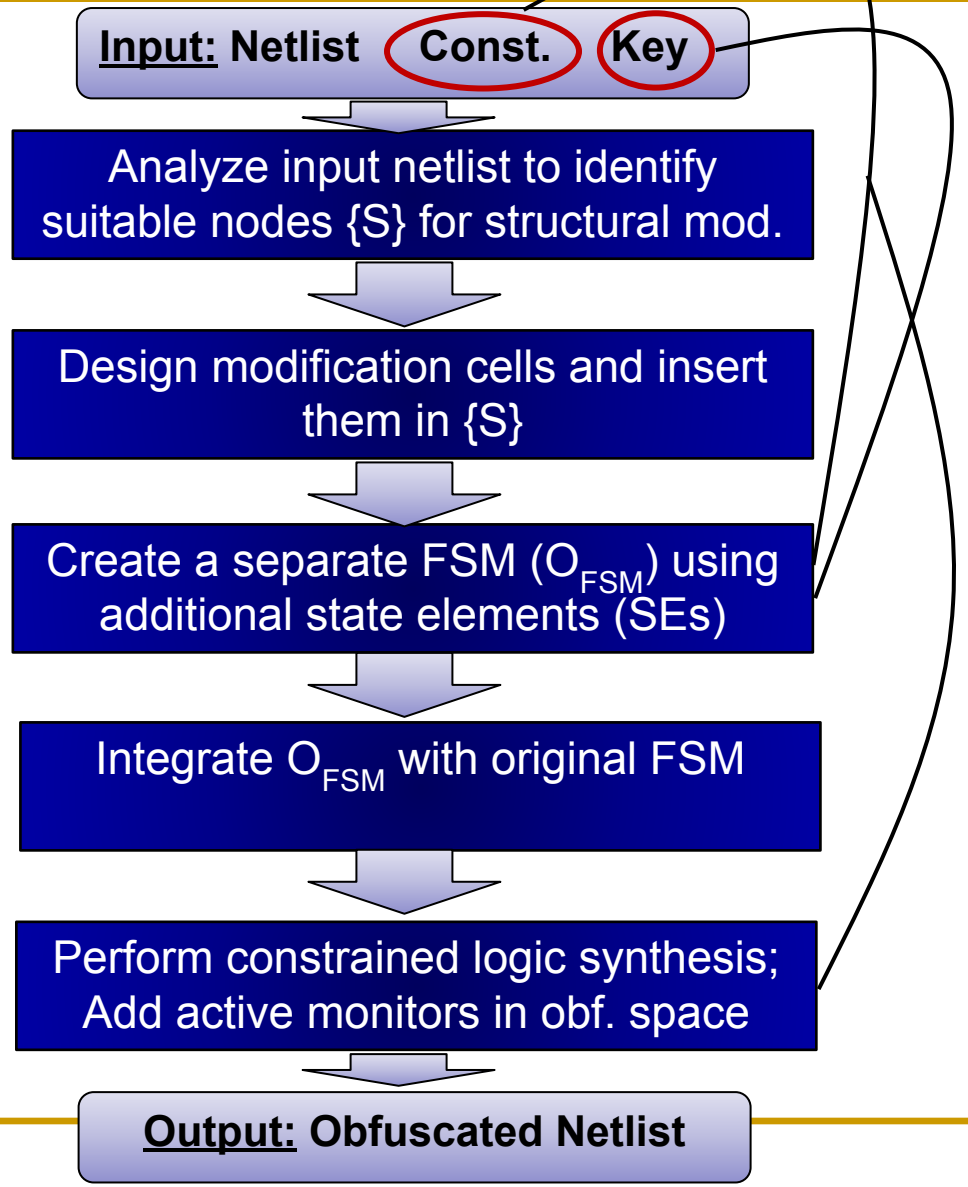
- It obfuscates the state space AND the comb. logic
- Uses rich theory of automata to transform the state space & associated logic



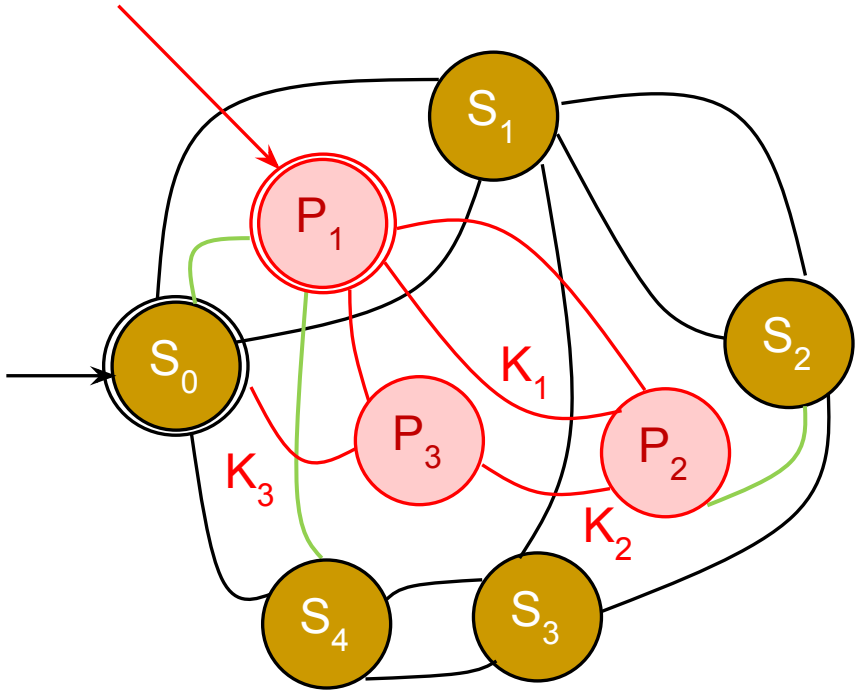


Protects against Piracy, RE & Tampering

The Flow



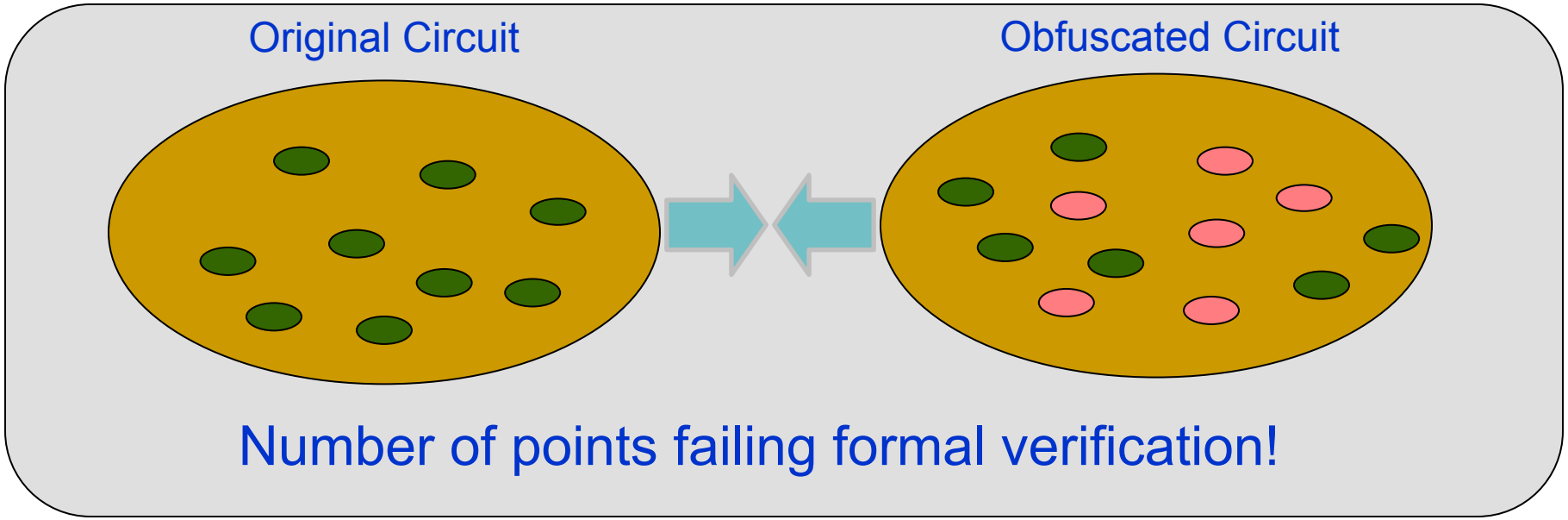
- Transforms underlying state machine



- Affects the dynamic behavior of the machine

Challenges

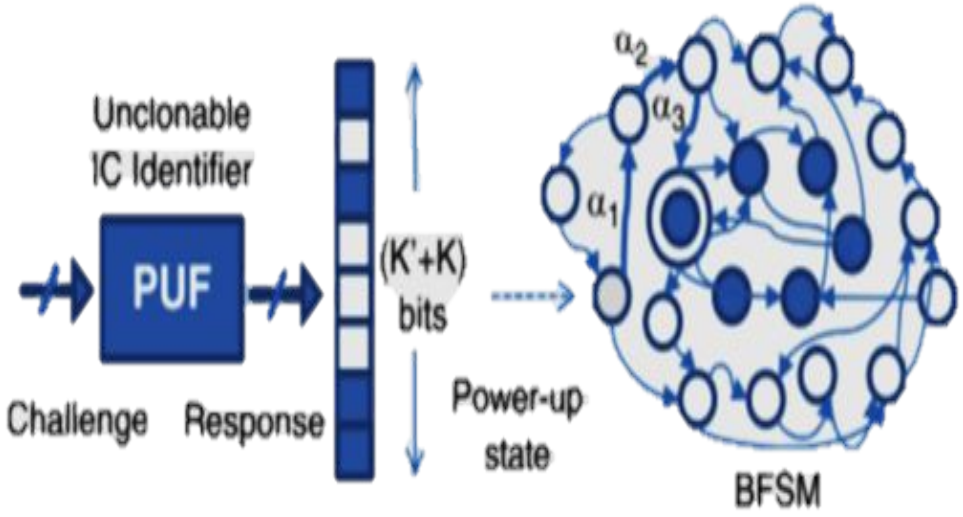
1. How to measure level of obfuscation?
2. How to measure the corresponding security benefit?



Improvement in *Trojan coverage* (w.r.t. defense against Trojan attacks)!

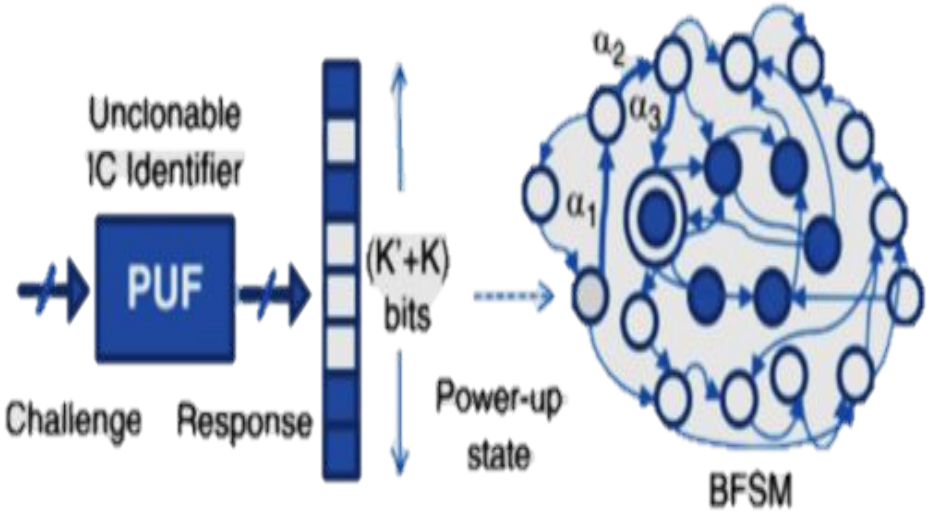
Internal (Integrated) Active Metering

- States and transitions for controlling chips are integrated within functional specifications
- $K = \log_2(S)$ flip flops needed to implement S states
- Adding S1 states requires K1=log(S1+S) flip flops
- Few additional flip flops can exponentially increase the number of states.

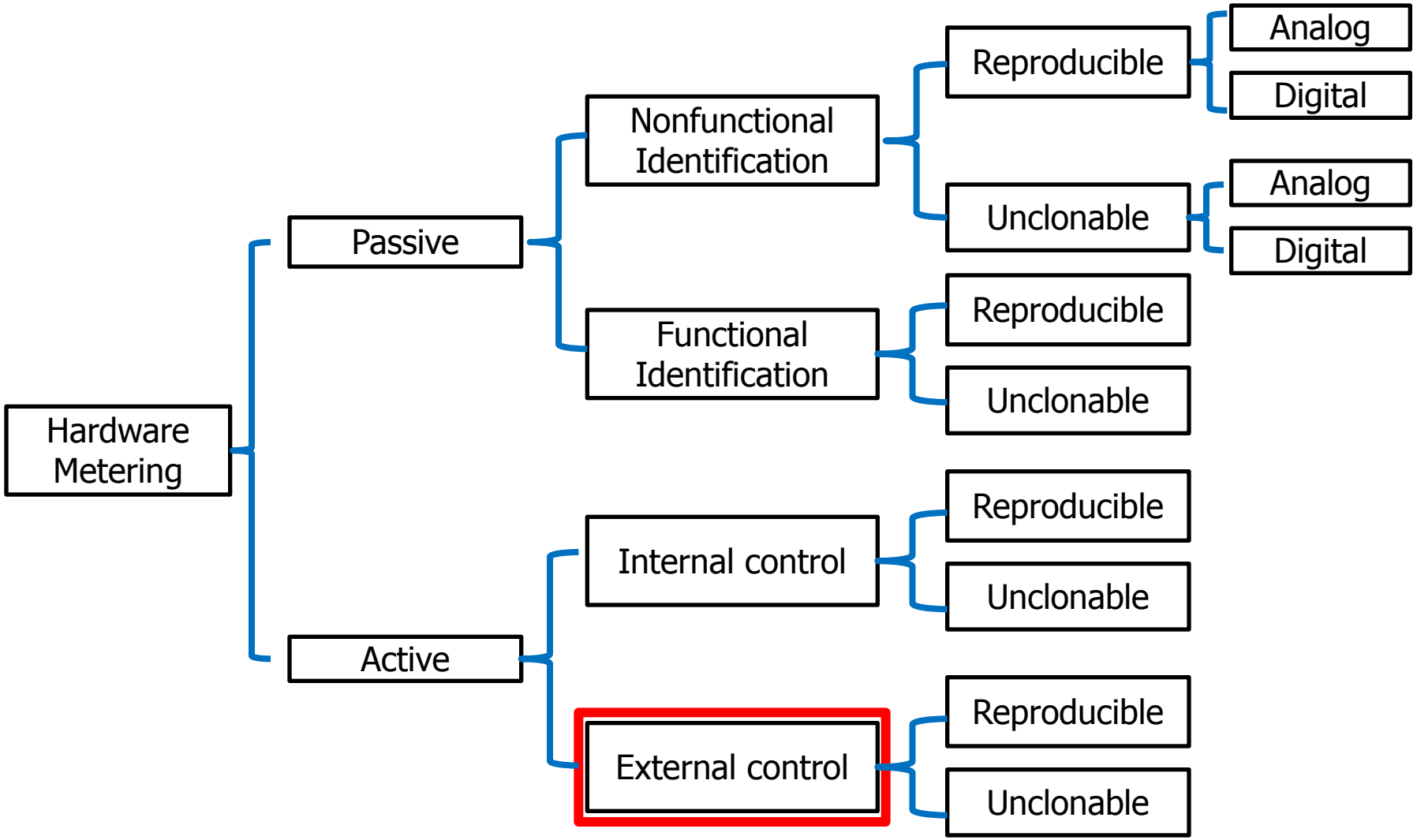


Internal (Integrated) Active Metering

- PUF generates random values, it sends device to random FSM state.
- Only IP owner with knowledge of FSM can find correct sequence to set FSM to reset state.
- Storing a sequence on chip requires additional logic such as clocks and memory and also requires chip to wait until entire sequence has been shifted in.



Taxonomy of Metering Methods

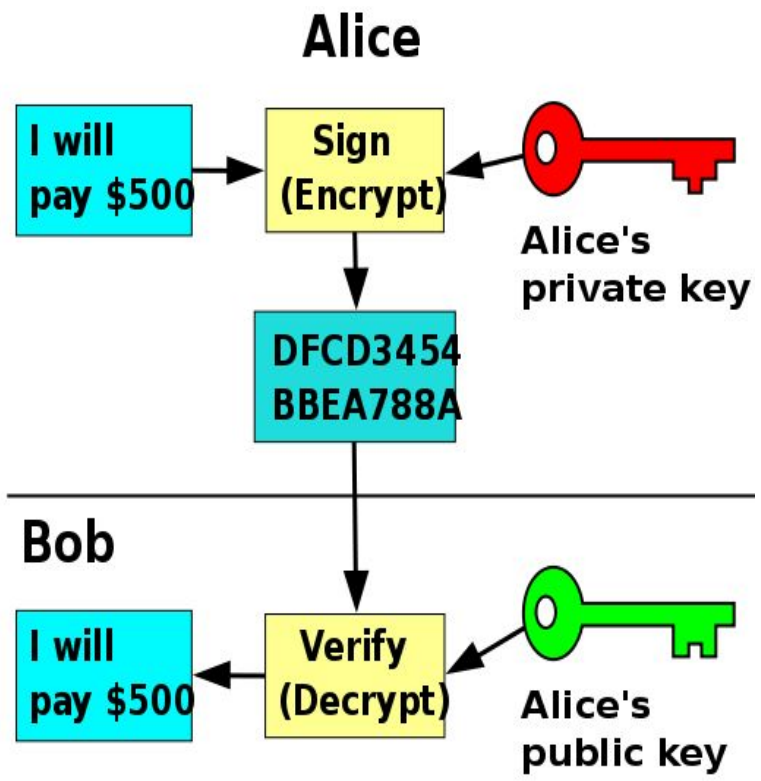


External Active Metering

- Uses external asymmetric cryptographic techniques to lock IC.
- Cryptographic circuits rely on public and private keys to give IP owner control over activation/correct function of the circuit.
- Only IP owner knows private key to unlock IC's functionality or testability.

Reminder: Public Key Cryptography

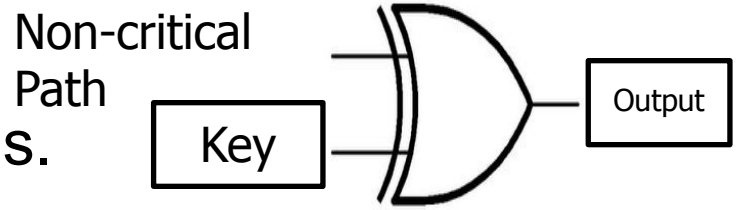
- Uses two large prime numbers p and q to generate co-prime n=pq
- Private (d) and public (e) keys based on n, p, and q are calculated
  - (e,n) are shared, message is encrypted using (d,n)
  - Decryption can be done using (e,n)
- Security relies on magnitude of prime numbers p and q



EPIC: Ending Piracy of Integrated Circuits

- This technique tries to allow IP Owner to have control over number of chips activated.

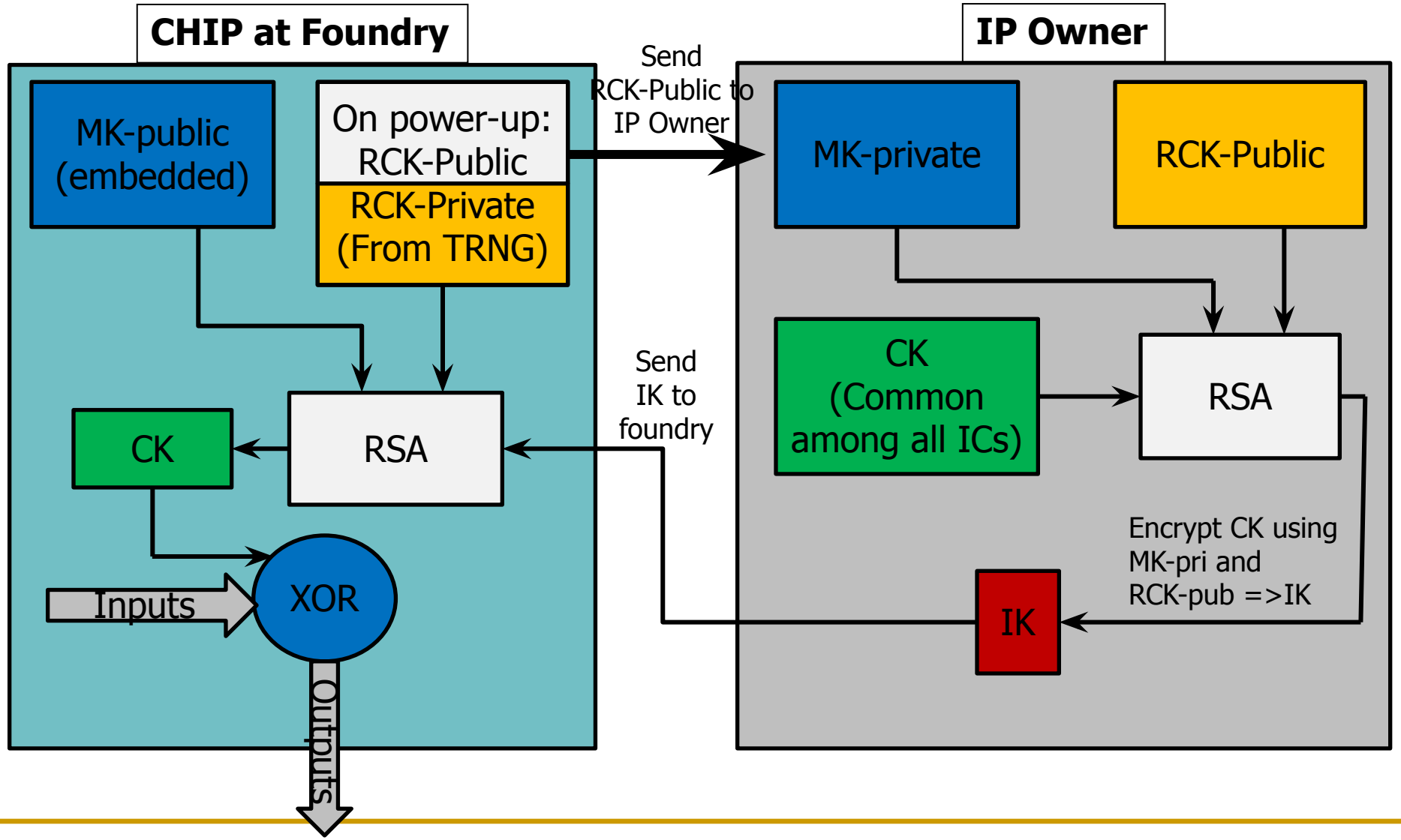
- Uses public-key encryption to lock correct functionality of chip.
- At the gate level, XOR gates are placed on selected non-critical paths.



- Requires that every chip be activated with an external key
  - Only IP owner can generate key

Roy et al., DATE 2008

EPIC High Level



EPIC

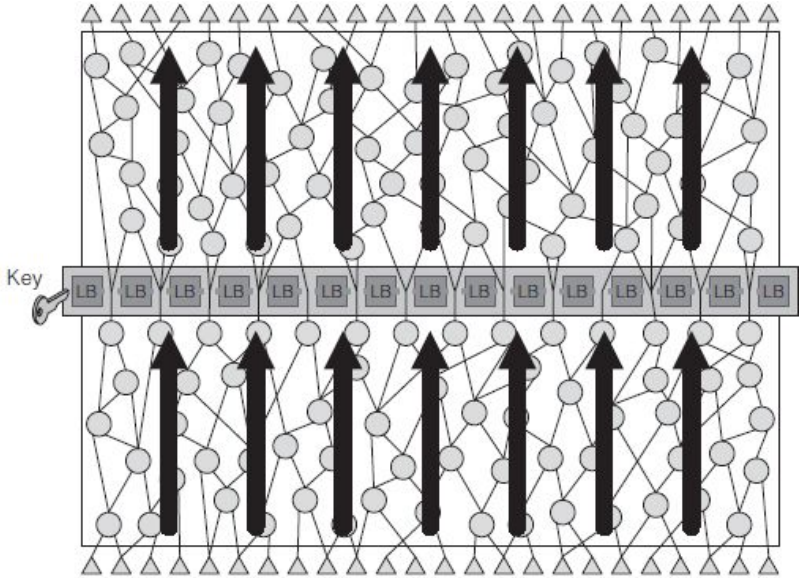
- Embedded in RTL is public Master Key (MK-Pub)
- XOR gates are controlled by Common Key. Correct Common Key unlocks circuit's correct functionality.
  - k-XOR gates need a common key of length k
- TRNG (True Random Number Generator) used to generate Random Chip Keys (RCK) on start up.
  - Upon power-up each chip generates a pair of private and public RCKs (RCK-private, RCK-public) which are **burned into programmable fuses.**
- Fab sends RCK-public to IP owner.

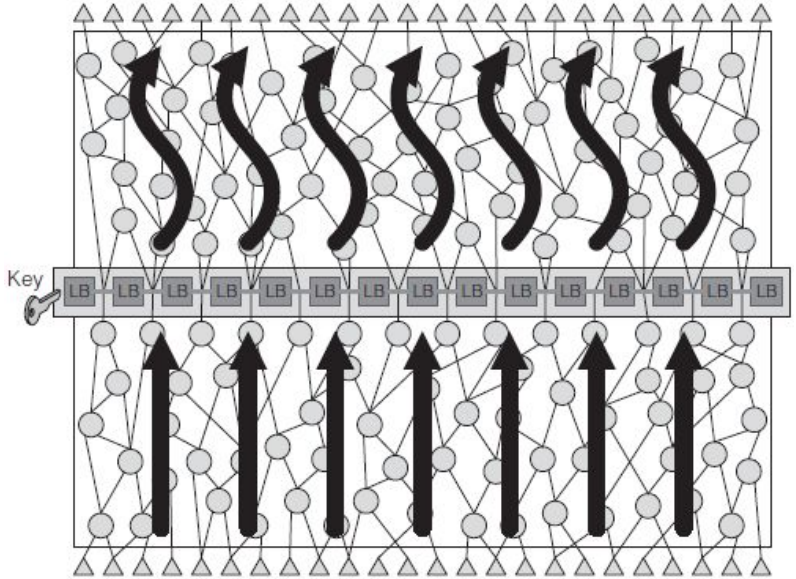
Analysis of EPIC

- Effective against cloned ICs.
  - Cloned ICs: Due to TRNG, each IC will have a unique random key, even cloned ICs. ICs need IK in order to be functional which only IP owner can generate.
- Not efficient against Over-produced ICs, Out-of-Spec ICs and defective ICs.
  - Over-produced ICs:
    - Fab could claim low yield and request more IKs than needed.
    - IP Owner has no way to verify yield or number of functional chips.
    - Foundry can still send keys to IP Owner. Keys are randomly generated and have no information on functionality of the IC.
  - Out-of-Spec ICs:
    - Foundry/assembly can send out the chip that are out of spec (their ID is a correct one)
  - Defective ICs:
    - Once IP owner sends Input Key, chip is activated. If chip is defective, IP Owner has no more communication with foundry and chip is already activated.

Reconfigurable Logic Barriers (LB)

- Separates inputs from outputs such that every path from input to output passes through a barrier.
- Logic barrier (LB) is a group of logic that allows correct path only if correct key is applied.





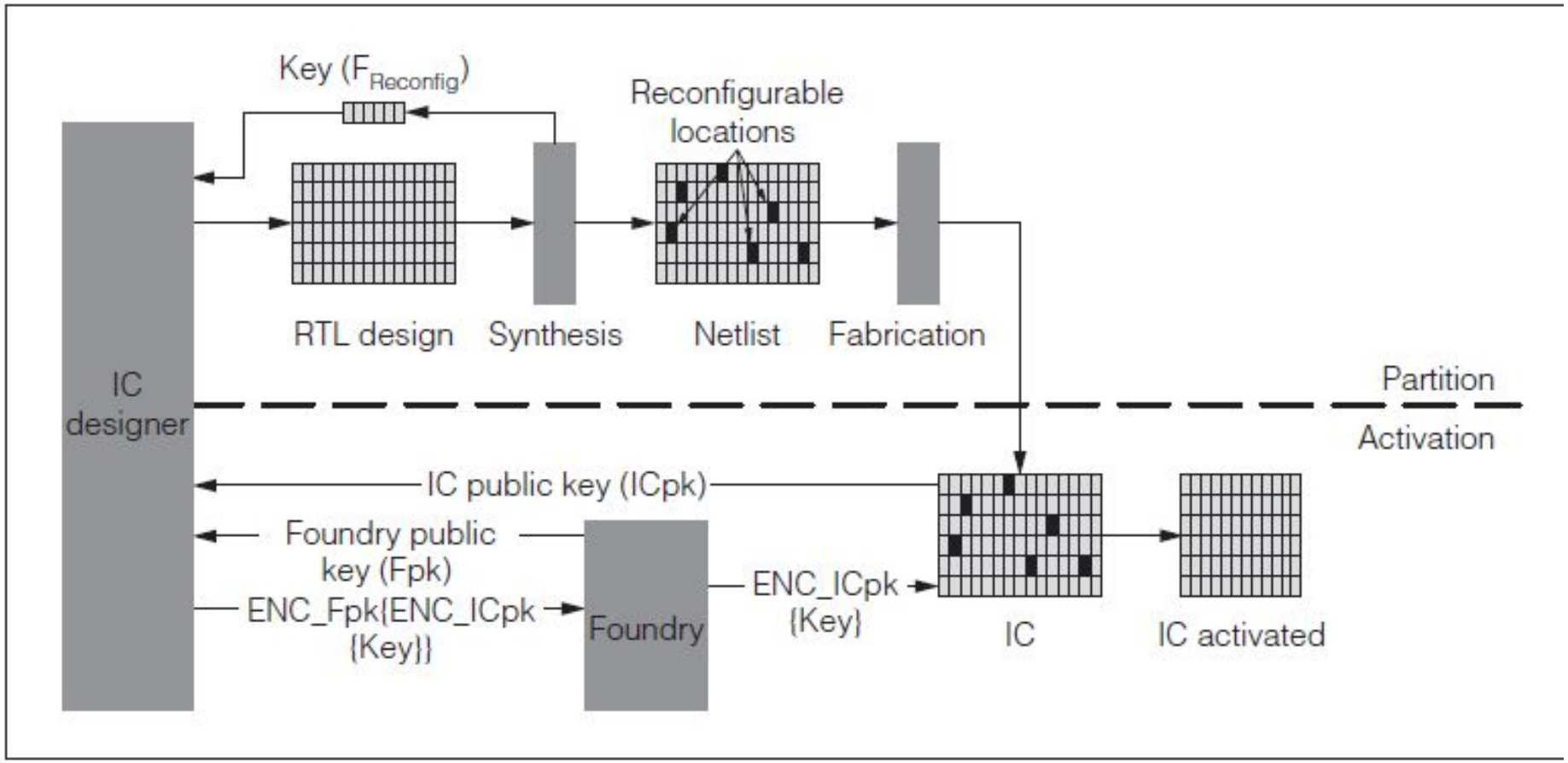
Reconfigurable Logic Barriers

- IP owner decomposes IC functionality into F_{fixed} and $F_{reconfig}$.
- F_{fixed} is given to foundry to fabricate.
- $F_{reconfig}$ is location of reconfigurable logic in combination with key needed to configure them correctly
- $F_{reconfig}$ can be programmed into reconfigurable locations using a secure key.

LB: Public Key Cryptography

- ICs use PUFs or TRNGs to generate a private and public random keys.
  - Public key from chip is sent to IP Owner
- IP Owner uses public key and its own private key to encrypt unlocking key.
  - Encrypted key is decrypted on chip using IP Owner's public key and chip's private key.

LB: Partitioning of Design

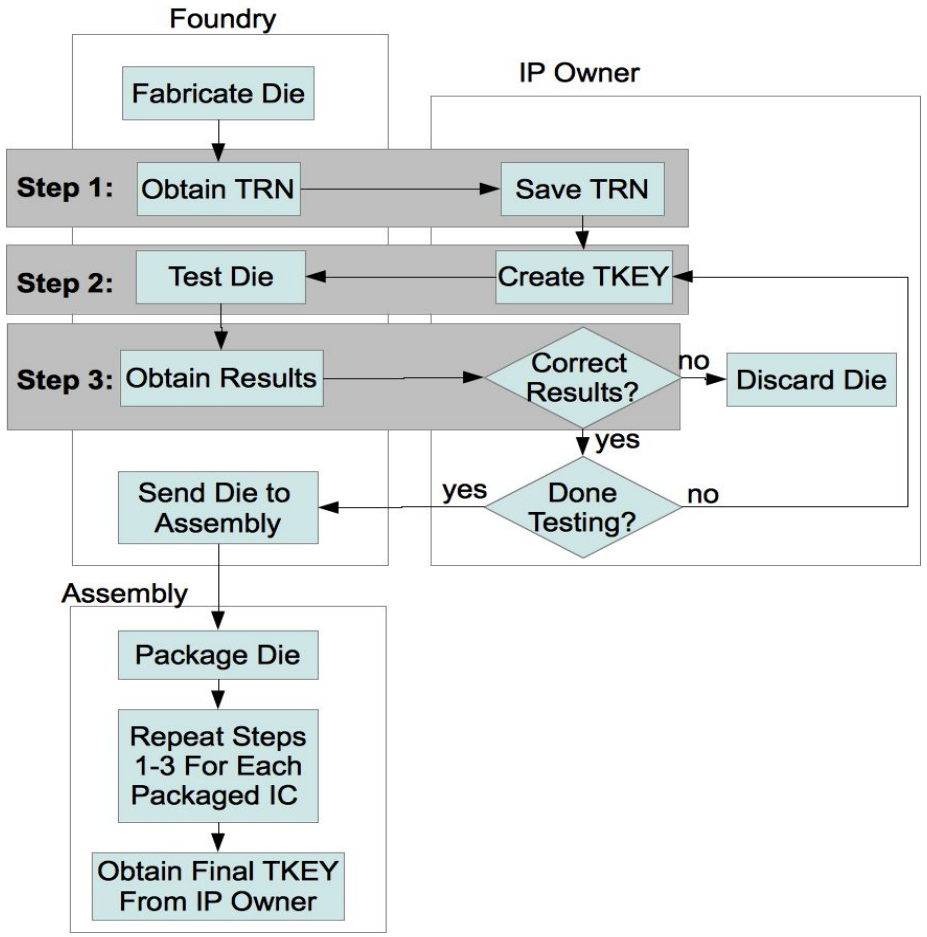


Logic Barriers Analysis

- **Effective against cloned ICs.**
  - Chips are only functional if correct key is entered which only IP Owner can provide
- **Ineffective against over-produced, defective, and out-of-spec ICs**
  - Foundry can lower yield in order to receive additional keys to activate functionality.
  - Key generated by chip does not have information about its functionality. Once key is applied, chip is functional.
- **Disadvantages:**
  - Look up tables require significant area overhead – 5X more than using XOR gates, and timing overhead.

Secure Split-Test (SST)

- Adds multiple layers of communication between IP owner, foundry, and assembly
- Ensures that IP owner will know exactly how many chips pass the test and how many have failed.
- Only chips that IP Owner has deemed functional will be given a functional key.



Test Seems to be a Challenge!

Designer

Test Patterns
Test Responses

Most techniques do not take into account the role “test” plays in the decision making process

Foundry & Assembly

Secure Split-Test

Designer

1. Designer has already put in hooks in the design that can ensure non-functional operation if the correct key is not included in the chip
2. Detecting a non-functional chip is significantly easier than using PUF and dealing with process variations

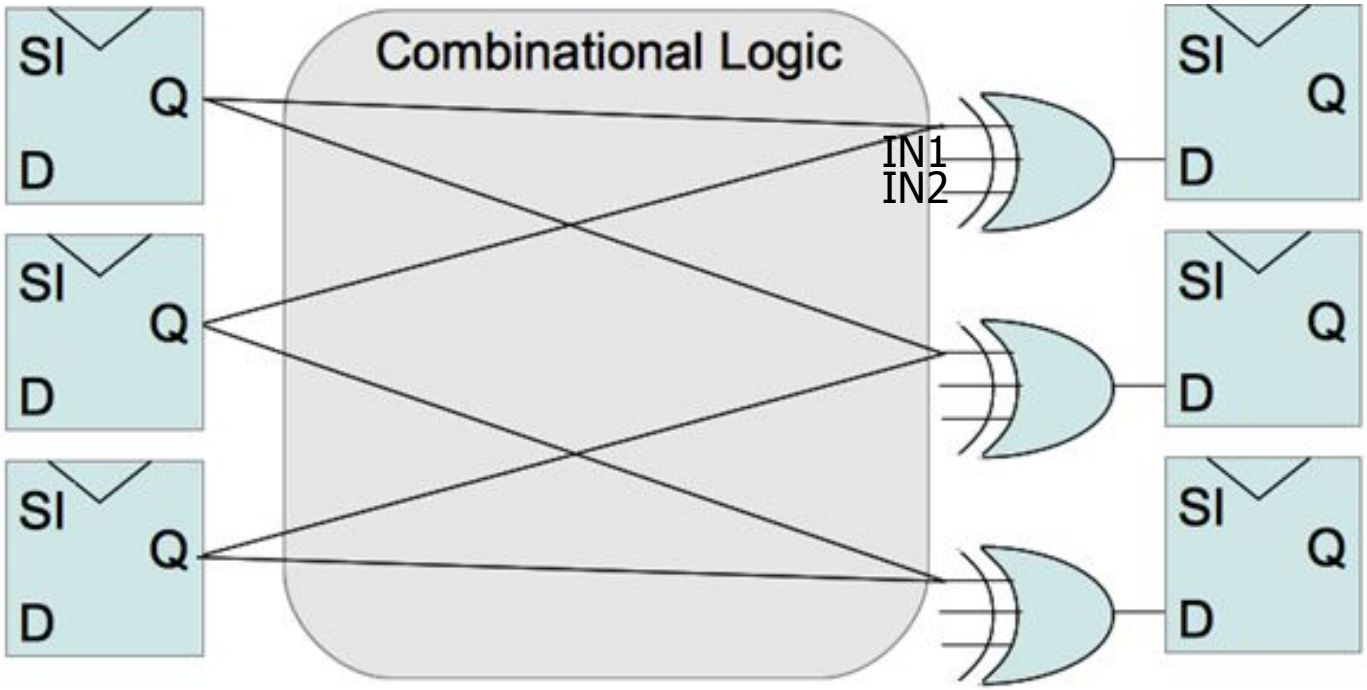
Secure Spilt Test

1. Foundry will not be able to ship any functional chips to the market
2. Same for defective chips and out-of-spec chips; the chips are simply non-functional.

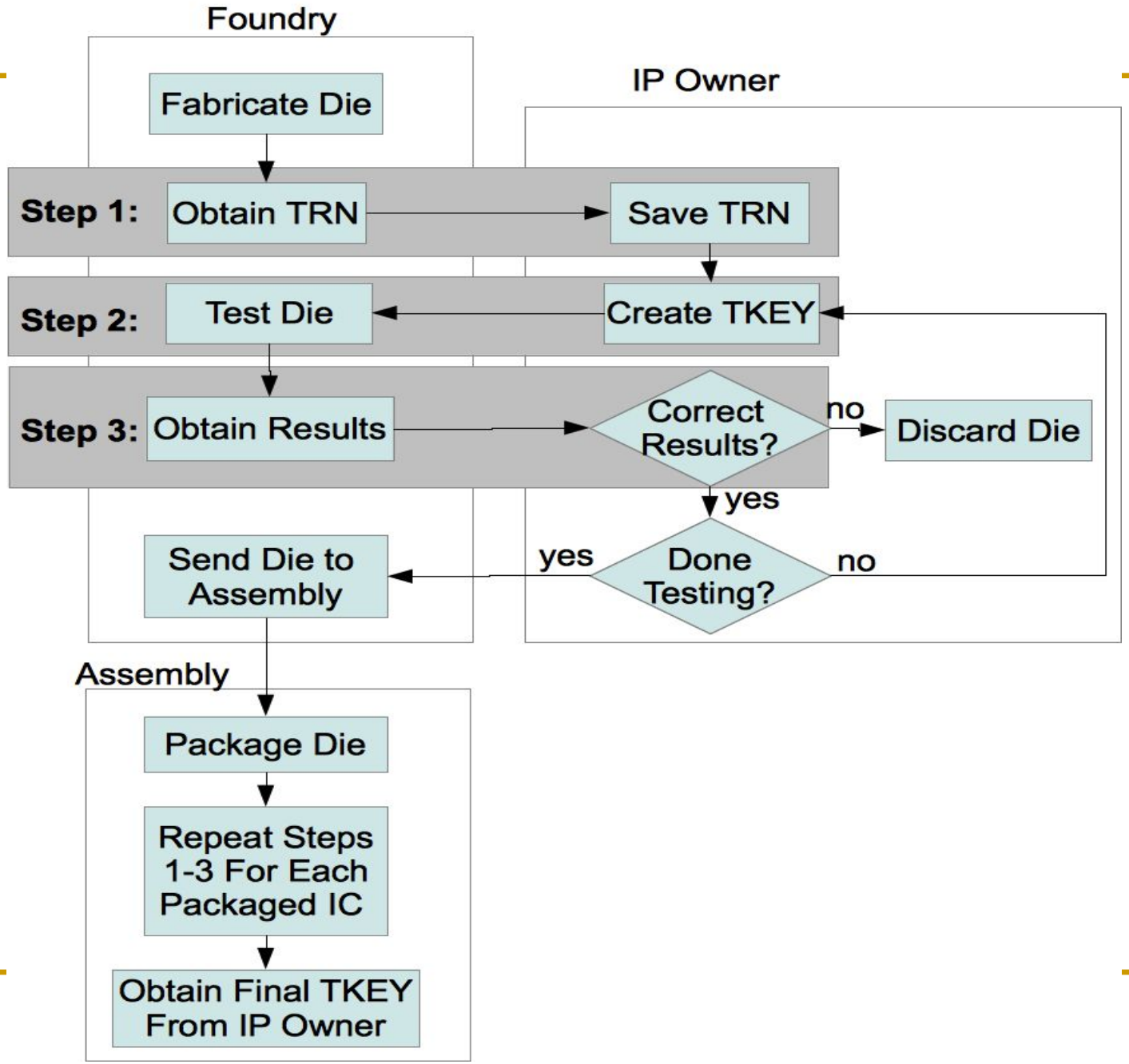
Foundry & Assembly

XOR Mask

- Three-input XOR logic added to non-critical paths.
- XOR logic additional inputs are IN1 and IN2



SST


Foundry
IP Owner
Fabricate Die
Step 1:
Obtain TRN
Save TRN
Step 2:
Test Die
Create TKEY
Step 3:
Obtain Results
Correct Results?
no
Discard Die
yes
Send Die to Assembly
yes
Done Testing?
no
Assembly
Package Die
Repeat Steps 1-3 For Each Packaged IC
Obtain Final TKEY From IP Owner

SST Analysis

- Effective against overproduced ICs, cloned ICs, and defective ICs
  - **Overproduced**:
    - IP Owner has control over number of TRNs received and TKEY/FKEYS sent to foundry/assembly
  - **Cloned**:
    - Chips are not functional unless FKEY has been produced by IP Owner
  - **Defective ICs:**
    - Foundry sends test results to foundry who checks results and decides if chip has correct test responses (chip is not yet functional at this stage)

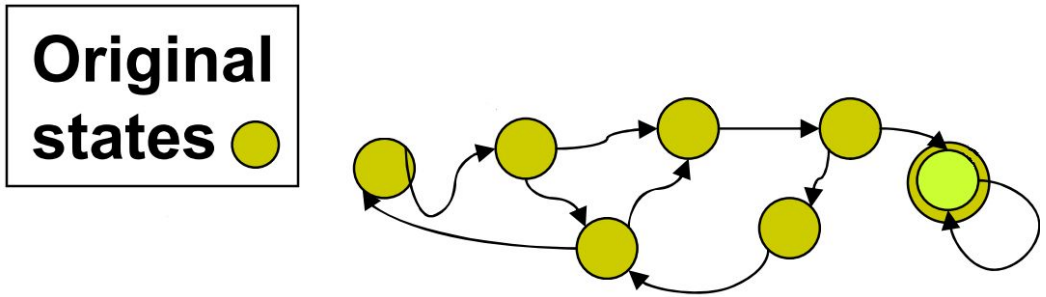
SST Analysis

- **Prevents out-of-spec ICs**
  - Some specifications cannot be determined from patterns testing alone. If a chip does not meet these specifications, it could be considered as a passing chip.
  - With the addition of a few sensors on the chip, these specifications can be tested and checked by IP Owner during SST
  - The IP owner will then be able to decide whether or not a chip passes the desired specifications in order to prevent out-of-spec ICs from going into market.

Remote Activation of ICs Through FSM Modification

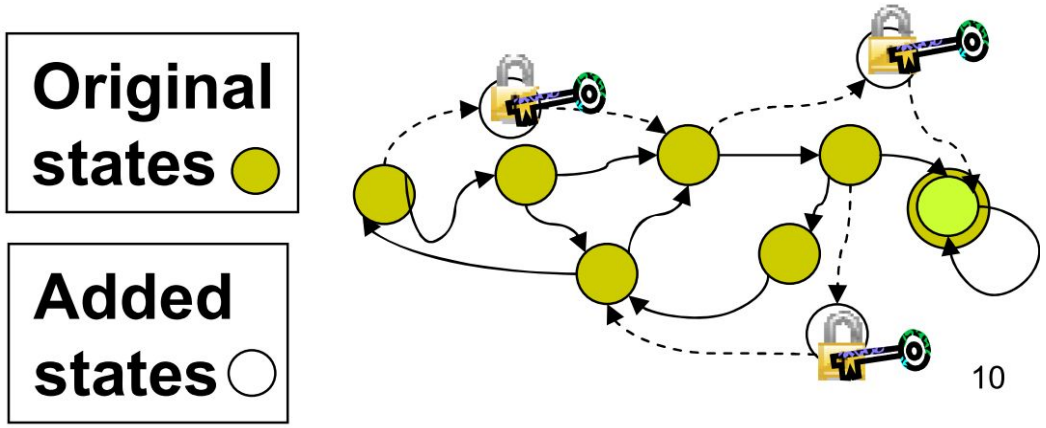
- FSM: Finite State Machine
- Sequence of inputs drive machine through different functional states
- Correct transitions give functional output

FSM



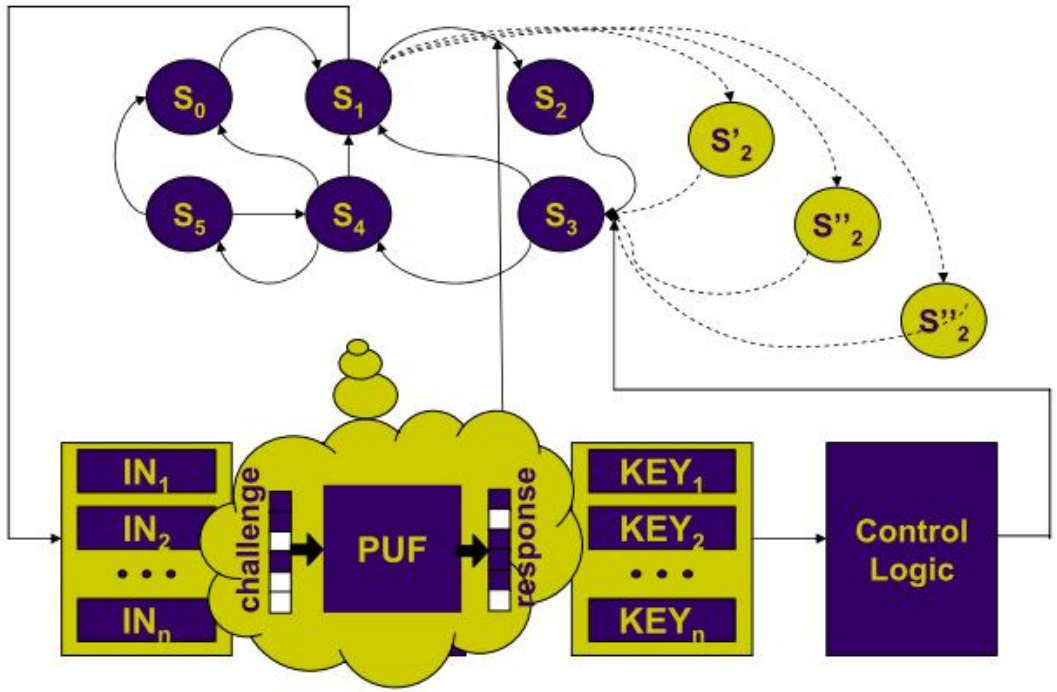
- Correct transitions give functional output
- Adding states to FSM gives IP owner controllability over sequence to reach functional states.

Boosted FSM (BFSM)



- On startup, inputs cause chip to go to one of added states
- IP Owner is only one with knowledge of FSM
- Only IP Owner knows right sequence (key) to bring FSM back to functional states.

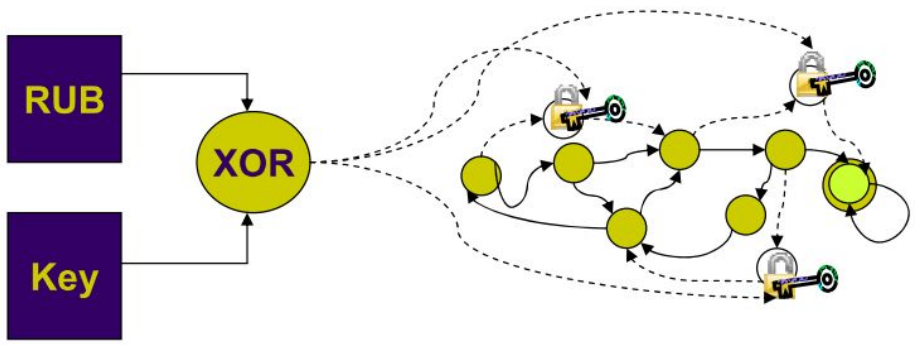
Remote Activation of ICs

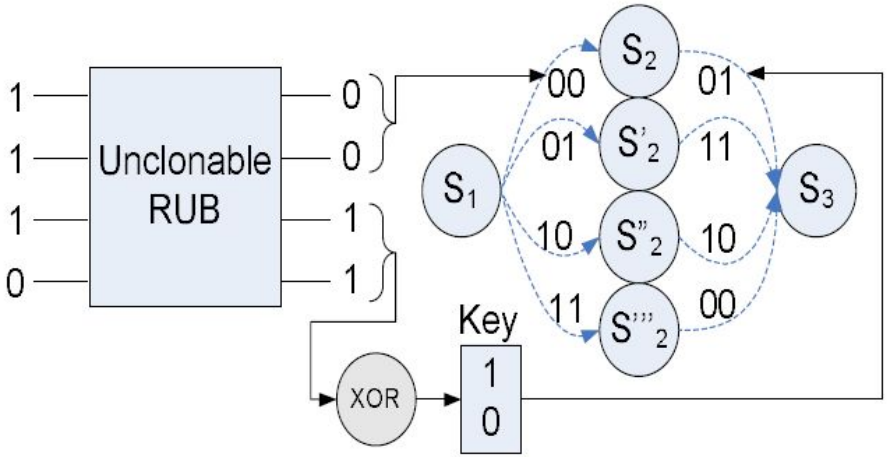


- Redundant states are added.
- Far less states needed than BFSM
- PUF response will send FSM to one of redundant states.

- **Challenge**: PUF is yet to be reliable.

Remote Activation of ICs





- RUB: Random Unique Block
- RUB must be stable – not change over time
- PUF (RUB) response is sent to IP Owner to generate key
- Key is then used to send FSM to correct state.

Analysis of Boosted FSM and Remote Activation

- BFSM requires many additional FSM states.
- Remote activation only uses a few redundant states.
- Both use PUF which is affected by age, temperature, noise, etc.
- Both effective against cloned ICs but not effective against defective, over-produced, or out-of-spec ICs.

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